



DC – AI – Routing – Switching – Security – Wireless – Automation

Apstra Device Replacement

An HPE Juniper Networking Techpost

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HPE



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1 Apstra Device Replacement

Bill Wester - 10/05/2023

An essential operation in a working data center network would be the need to replace a device that has either failed or just needs to be re-allocated/reused for other purposes. This document describes the steps needed to accomplish this task via the Apstra UI and the Apstra Terraform Provider.

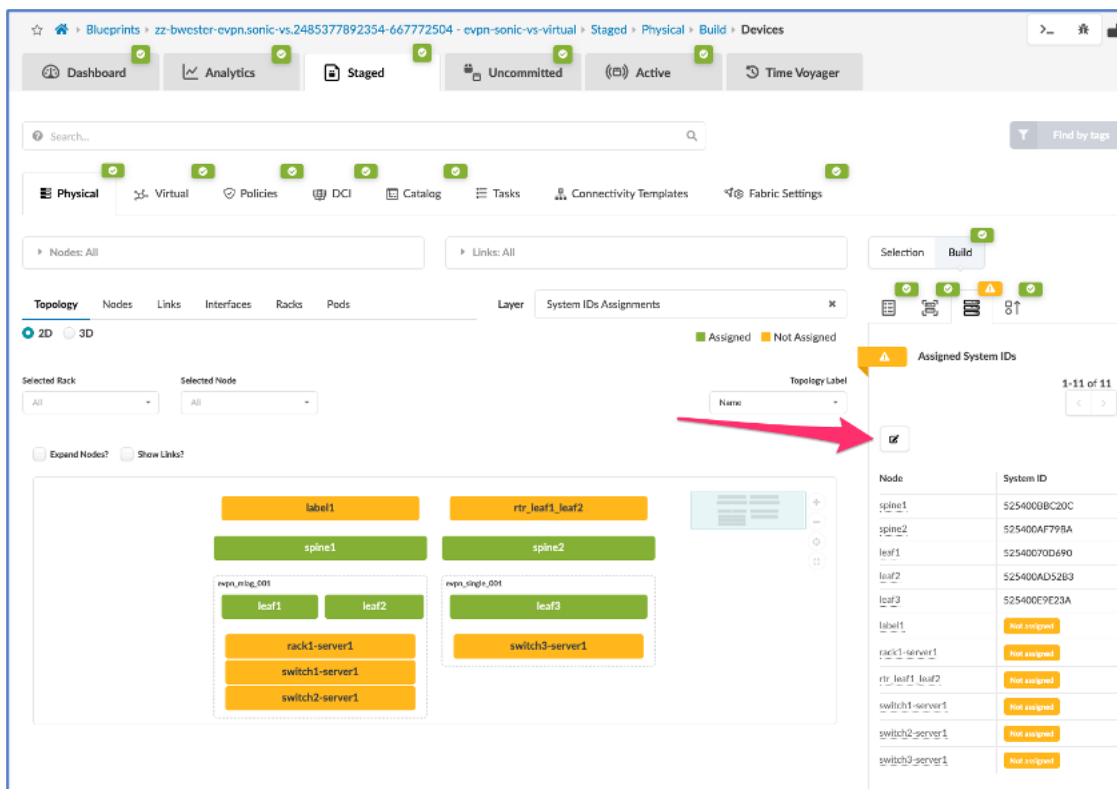
Note that we did not have physical hardware to perform this operation in our test environment.

1.1 High Level Steps Required

Conceptually replacing a device for any reason in Apstra is a straight forward set of tasks. Devices are abstracted from their role in the fabric by the use of Logical Devices and Interface Maps and are assigned to a blueprint. In this use case we are expecting the replacement device to be exactly the same physically (ie. make, model and port configuration) as the replaced device.

1.1.1 GUI Method Deletion of leaf2 with Still Connected Device

1. Navigate via the Apstra UI to Blueprints -> blueprint_ID -> staged -> Physical -> Build -> Devices and use the edit pencil to “Change system ID Assignments”



2. We will choose to work with the leaf2 device. Delete the leaf2 assignment by selecting the trashcan icon.
3. Now choose the 'Update Assignment' selection.

Assign Systems

Query: All 1-11 of 11

Name	Role	Hostname	System ID	Deploy Mode
spine1	Spine	spine1	525400BBC20C (10.28.24.11) - spine1	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
spine2	Spine	spine2	525400AF79BA (10.28.24.12) - spine2	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
leaf1	Leaf	leaf1	52540070D690 (10.28.24.13) - leaf1	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
leaf2	Leaf	leaf2	525400AD52B3 (10.28.24.14) - leaf2	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
leaf3	Leaf	leaf3	525400E9E23A (10.28.24.15) - leaf3	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
label1	Generic System	hostname1	Select...	☐ Deploy ☐ Ready ☐ Undeploy
rack1-server1	Generic System	rack1-server1	Select...	☐ Deploy ☐ Ready

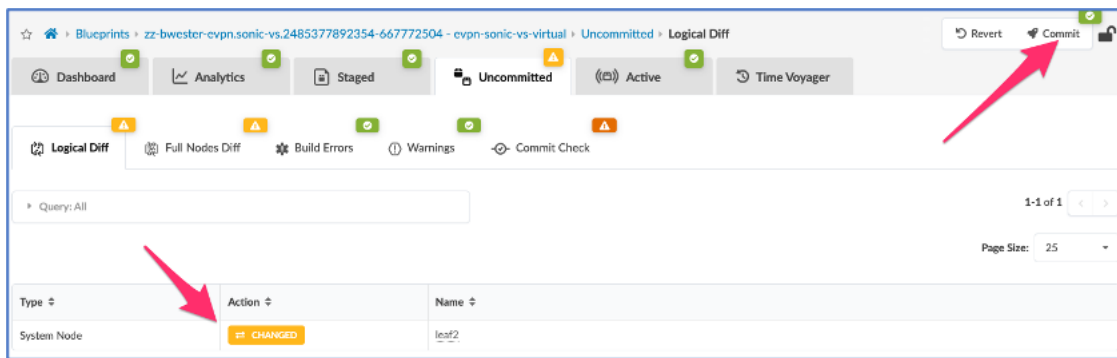
Update Assignments

Here we can see the leaf2 is no longer assigned:

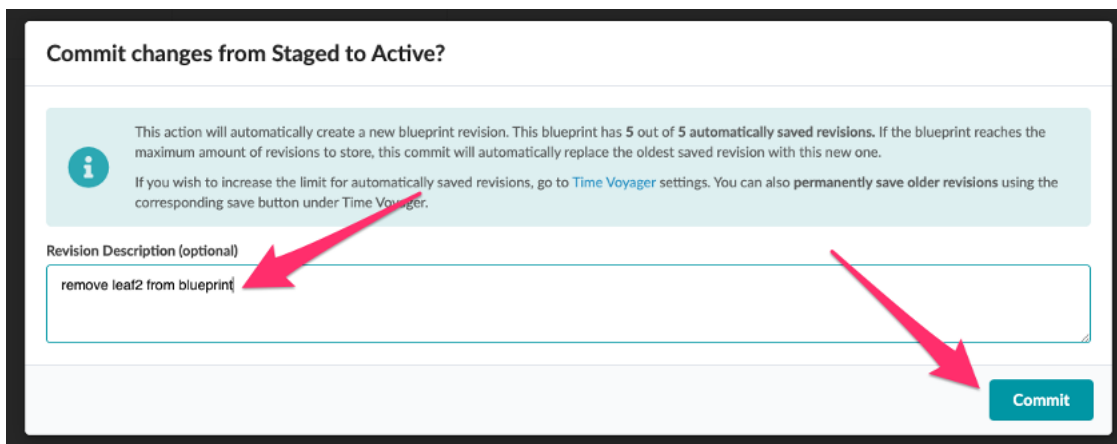
The screenshot shows the 'Assigned System IDs' section in the Apstra interface. It features a table with two columns: 'Node' and 'System ID'. The table lists 11 nodes, with their system IDs or 'Not assigned' status. A red arrow points to the 'Not assigned' status for 'leaf2'.

Node	System ID
spine1	525400BBC20C
spine2	525400AF79BA
leaf1	52540070D690
leaf2	Not assigned
leaf3	525400E9E23A
label1	Not assigned
rack1-server1	Not assigned
rtr_leaf1_leaf2	Not assigned
switch1-server1	Not assigned
switch2-server1	Not assigned
switch3-server1	Not assigned

4. Commit the change to the system, navigate to Blueprint-> blueprintID -> Uncommitted -> and click the "Commit" rocket icon:



You should then add a comment and complete the commit:



5. Once the commit is complete, navigate to Devices -> Managed Devices and you should now see that leaf2 is unassigned from the blueprint:

The screenshot shows the 'Managed Devices' table in the Apstra interface. The table has columns for 'Management IP', 'Device Key', 'Device Profile', 'Hostname', 'OS', 'State', 'Comms', 'Acknowledged?', 'Blueprint', 'Type', 'Agent Profile', 'Apstra Version', 'Last Job Type', 'Job State', and 'Actions'. A red arrow points to the 'Blueprint' column for the device with IP 10.28.24.14, which is 'Not assigned'.

Management IP	Device Key	Device Profile	Hostname	OS	State	Comms	Acknowledged?	Blueprint	Type	Agent Profile	Apstra Version	Last Job Type	Job State	Actions
10.28.24.11	525409B8C20C	VS_SONIC_BUZZNIK_PLUS	spine1	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bwester-evpn-sonic-vs-2485377892354-667772504 - evpn-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮
10.28.24.15	525409F9E23A	VS_SONIC_BUZZNIK_PLUS	leaf3	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bwester-evpn-sonic-vs-2485377892354-667772504 - evpn-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮
10.28.24.14	525409AD52B3	VS_SONIC_BUZZNIK_PLUS	leaf2	SONIC 4.0.5-Enterprise_Base	OOS-READY	✓	✓	Not assigned	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮
10.28.24.12	525409AF796A	VS_SONIC_BUZZNIK_PLUS	spine2	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bwester-evpn-sonic-vs-2485377892354-667772504 - evpn-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮
10.28.24.13	52540970D690	VS_SONIC_BUZZNIK_PLUS	leaf1	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bwester-evpn-sonic-vs-2485377892354-667772504 - evpn-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮

6. Now we can uninstall the agent from the device. Navigate to Devices -> Managed devices and you should be able to select the correct device (10.28.24.14 in this example) Check the box next to the device and then click the uninstall icon:

The screenshot shows the 'Managed Devices' page in Apstra. At the top, there are buttons for 'Create Onbox Agent(s)', 'Create Offbox Agent(s)', and 'Advanced Settings'. Below these is a search bar and a filter section. A red arrow points to the 'Uninstall' button in the top toolbar. The main table lists devices with columns for Management, Device Key, Device Profile, Hostname, OS, State, Comms, Acknowledged, Blueprint, Type, Agent Profile, Apstra Version, Last Job Type, Job State, and Actions. The device 'sonic' (10.28.24.14) is selected, and its 'Uninstall' button is highlighted.

Management	Device Key	Device Profile	Hostname	OS	State	Comms	Acknowledged	Blueprint	Type	Agent Profile	Apstra Version	Last Job Type	Job State	Actions
10.28.24.11	S254008BC20C	VS_SONIC_BUZZNIK_PLUS	spine1	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bwester-eyon-sonic-v1.2485377892354-667772504-eyon-sonic-v1-virtual	ONBOX	UNASSIGNED	AO5_4.2.0_OB.232	INSTALL	SUCCESS	
10.28.24.13	S254008BC20C	VS_SONIC_BUZZNIK_PLUS	leaf3	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bwester-eyon-sonic-v1.2485377892354-667772504-eyon-sonic-v1-virtual	ONBOX	UNASSIGNED	AO5_4.2.0_OB.232	INSTALL	SUCCESS	
10.28.24.14	S25400AD52B3	VS_SONIC_BUZZNIK_PLUS	sonic	SONIC 4.0.5-Enterprise_Base	OOS-READY	✓	✓	Not assigned	ONBOX	UNASSIGNED	AO5_4.2.0_OB.232	INSTALL	SUCCESS	
10.28.24.12	S25400AF798A	VS_SONIC_BUZZNIK_PLUS	spine2	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bwester-eyon-sonic-v1.2485377892354-667772504-eyon-sonic-v1-virtual	ONBOX	UNASSIGNED	AO5_4.2.0_OB.232	INSTALL	SUCCESS	
10.28.24.13	S254007D3060	VS_SONIC_BUZZNIK_PLUS	leaf1	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bwester-eyon-sonic-v1.2485377892354-667772504-eyon-sonic-v1-virtual	ONBOX	UNASSIGNED	AO5_4.2.0_OB.232	INSTALL	SUCCESS	

This then should generate another popup confirmation:

The screenshot shows a confirmation dialog titled 'Uninstall selected elements'. It contains a table with columns: Device Address, Type, Agent Profile, Operation Mode, Platform, Platform Version, State, Job State, Connection State, System ID, Hostname, Device State, and Action Status. The device 'sonic' (10.28.24.14) is listed with a 'SUCCESS' state and 'UNAVAILABLE' connection state. A red arrow points to the 'Uninstall selected elements' button at the bottom right.

Device Address	Type	Agent Profile	Operation Mode	Platform	Platform Version	State	Job State	Connection State	System ID	Hostname	Device State	Action Status
10.28.24.14	ONBOX	UNASSIGNED	FULL CONTROL	SONIC	SONIC-OS-4.0.5-Enterprise_Base	SUCCESS		UNAVAILABLE	S25400AD52B3	sonic	OOS-READY	N/A

7. Once this task completes you should observe the state for leaf2 is oos-nocomms and a red x in the 'comms' column:

The screenshot shows the 'Managed Devices' page after the uninstallation. The device 'leaf2' (10.28.24.13) now has a state of 'OOS-NOCOMMS' and a red 'x' in the 'Comms' column. A red arrow points to the 'Comms' column for this device.

Management	Device Key	Device Profile	Hostname	OS	State	Comms	Acknowledged	Blueprint	Type	Agent Profile	Apstra Version	Last Job Type	Job State	Actions
10.28.24.11	S254008BC20C	VS_SONIC_BUZZNIK_PLUS	spine2	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bwester-eyon-sonic-v1.2485377892354-667772504-eyon-sonic-v1-virtual	ONBOX	UNASSIGNED	AO5_4.2.0_OB.232	INSTALL	SUCCESS	
10.28.24.13	S254008BC20C	VS_SONIC_BUZZNIK_PLUS	leaf3	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bwester-eyon-sonic-v1.2485377892354-667772504-eyon-sonic-v1-virtual	ONBOX	UNASSIGNED	AO5_4.2.0_OB.232	INSTALL	SUCCESS	
10.28.24.14	S25400AD52B3	VS_SONIC_BUZZNIK_PLUS	sonic	SONIC 4.0.5-Enterprise_Base	OOS-NOCOMMS	✗	✓	Not assigned	ONBOX	UNASSIGNED	AO5_4.2.0_OB.232	UNINSTALL	SUCCESS	
10.28.24.12	S25400AF798A	VS_SONIC_BUZZNIK_PLUS	spine2	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bwester-eyon-sonic-v1.2485377892354-667772504-eyon-sonic-v1-virtual	ONBOX	UNASSIGNED	AO5_4.2.0_OB.232	INSTALL	SUCCESS	

8. Delete the device completely from Apstra by selecting the device and clicking the trashcan icon:



The screenshot shows the Apstra GUI interface. The top navigation bar includes 'Blueprints', 'zz-bwester-evpn-sonic-vs.2485377892354-667772504', 'evpn-sonic-vs-virtual', 'Staged', 'Physical', 'Build', and 'Devices'. The main menu has tabs for 'Physical', 'Virtual', 'Policies', 'DCI', 'Catalog', 'Tasks', 'Connectivity Templates', and 'Fabric Settings'. The 'Physical' tab is active, showing a topology diagram with nodes like 'label1', 'spine1', 'spine2', 'leaf1', 'leaf2', 'leaf3', 'rack1-server1', 'switch1-server1', 'switch2-server1', and 'switch3-server1'. The 'Assigned System IDs' panel on the right lists the following data:

Node	System ID
spine1	S25400B0C20C
spine2	S25400AF798A
leaf1	S2540070D690
leaf2	S25400AD52B3
leaf3	S25400E9E23A
label1	Not assigned
rack1-server1	Not assigned
rt_r_leaf1_leaf2	Not assigned
switch1-server1	Not assigned
switch2-server1	Not assigned
switch3-server1	Not assigned

2. Delete the leaf2 assignment by selecting the trashcan icon.
3. Now choose the 'Update Assignment' selection.

Assign Systems

Query: All 1-11 of 11

Name	Role	Hostname	System ID	Deploy Mode
spine1	Spine	spine1	525400BBC20C (10.28.24.11) - spine1	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
spine2	Spine	spine2	525400AF79BA (10.28.24.12) - spine2	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
leaf1	Leaf	leaf1	52540070D690 (10.28.24.13) - leaf1	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
leaf2	Leaf	leaf2	525400AD52B3 (10.28.24.14) - leaf2	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
leaf3	Leaf	leaf3	525400E9E23A (10.28.24.15) - leaf3	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
label1	Generic System	hostname1	Select...	☐ Deploy ☐ Ready ☐ Undeploy
rack1-server1	Generic System	rack1-server1	Select...	☐ Deploy ☐ Ready

Update Assignments

4. Commit the change to the system, navigate to Blueprint-> blueprintID -> Uncommitted -> and click the “Commit” rocket icon:

☆ > Blueprints > zz-bwester-cvpn,sonic-vs,2485377892354-667772504 - cvpn-sonic-vs-virtual > Uncommitted > Logical Diff

Dashboard Analytics Staged Uncommitted Active Time Voyager

Logical Diff Full Nodes Diff Build Errors Warnings Commit Check

Query: All 1-1 of 1 Page Size: 25

Type	Action	Name
System Node	CHANGED	leaf2

You should then add a comment and complete the commit:

Commit changes from Staged to Active?

This action will automatically create a new blueprint revision. This blueprint has 5 out of 5 automatically saved revisions. If the blueprint reaches the maximum amount of revisions to store, this commit will automatically replace the oldest saved revision with this new one.

If you wish to increase the limit for automatically saved revisions, go to [Time Voyager](#) settings. You can also permanently save older revisions using the corresponding save button under Time Voyager.

Revision Description (optional)

remove leaf2 from blueprint

Commit

5. Once the commit is complete, navigate to Devices -> Managed Devices and you should now see that leaf2 is unassigned from the blueprint:

Devices > Managed Devices

Query: All 1-5 of 5 Columns (15/17) Page Size: 25

Filter selected by ☒ all ☐ selected only ☐ unselected only

Device Information										Agent Information				
Management IP	Device Key	Device Profile	Hostname	OS	State	Comms	Acknowledged?	Blueprint	Type	Agent Profile	Apstra Version	Last Job Type	Job State	Actions
10.28.24.11	5254088C28C	VS_SONIC_BUZZNIK_PLUS	spine1	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bowster-cvxn-sonic-vs-2485377892354-667772504-cvxn-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	
10.28.24.15	5254089F23A	VS_SONIC_BUZZNIK_PLUS	leaf3	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bowster-cvxn-sonic-vs-2485377892354-667772504-cvxn-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	
10.28.24.14	525408AD52B3	VS_SONIC_BUZZNIK_PLUS	leaf2	SONIC 4.0.5-Enterprise_Base	OOS-REACTIV	✓	✓	Not assigned	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	
10.28.24.12	525408AF796A	VS_SONIC_BUZZNIK_PLUS	spine2	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bowster-cvxn-sonic-vs-2485377892354-667772504-cvxn-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	
10.28.24.13	5254070D690	VS_SONIC_BUZZNIK_PLUS	leaf1	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	✓	✓	zz-bowster-cvxn-sonic-vs-2485377892354-667772504-cvxn-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	

6. Remove the failed leaf2 device from the Managed Devices list.

☆ 🏠 » Devices » Managed Devices

[+ Create Onbox Agent\(s\)](#)
[+ Create Offbox Agent\(s\)](#)
[Advanced Settings](#)

Query: All 1-6 of 6

Columns (15/17) Page Size: 25

Filter selected by ☒ all ☐ selected only ☐ unselected only

Device Information								Agent Information						
Management IP	Device Key	Device Profile	Hostname	OS	State	Comms	Acknowledged	Blueprint	Type	Agent Profile	Apstra Version	Last Job Type	Job State	Actions
10.28.24.11	5254008B20C	V5_SONIC_BUZZNIK_PLUS	spine1	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	🟢	🟢	22-bvster-evsn-sonic-vs-2485277892354-667772504-exp-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮
10.28.24.15	525400E9E23A	V5_SONIC_BUZZNIK_PLUS	leaf3	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	🟢	🟢	22-bvster-evsn-sonic-vs-2485277892354-667772504-exp-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮
10.28.24.12	525400AF79BA	V5_SONIC_BUZZNIK_PLUS	spine2	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	🟢	🟢	22-bvster-evsn-sonic-vs-2485277892354-667772504-exp-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮
10.28.24.13	525400700690	V5_SONIC_BUZZNIK_PLUS	leaf1	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	🟢	🟢	22-bvster-evsn-sonic-vs-2485277892354-667772504-exp-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮
10.28.24.14	525400A05283	V5_SONIC_BUZZNIK_PLUS	leaf2	SONIC 4.0.5-Enterprise_Base	OOS-NOCOMMS	🔴	🟢	Not assigned	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮
10.28.24.224	525400A052AA	V5_SONIC_BUZZNIK_PLUS		SONIC 4.0.5-Enterprise_Base	OOS-QUARANTINED	🔴	🔴	Not assigned	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮

Device Agent

🗑️

7. Confirm the deletion dialog.

Delete this resource?

🗑️ Delete

8. Remove the agent from the failed leaf2 device.

☆ > Devices > Managed Devices

Create Onbox Agent(s) Create Offbox Agent(s) Advanced Settings

Query: All 1-6 of 6 Columns (15/17) Page Size: 25

Filter selected by ☒ all ☐ selected only ☐ unselected only

Device Information										Agent Information					
Management IP	Device Key	Device Profile	Hostname	OS	State	Comms	Acknowledged?	Blueprint	Type	Agent Profile	Apstra Version	Last Job Type	Job State	Actions	
10.28.24.11	525400B6C20C	VS_SONIC_BUZZNIK_PLUS	spine1	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	🟢	🟢	22-bwester-evps@sonic-vs.2485377892354-6677722504 - evps-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮	
10.28.24.15	525400E9E23A	VS_SONIC_BUZZNIK_PLUS	leaf3	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	🟢	🟢	22-bwester-evps@sonic-vs.2485377892354-6677722504 - evps-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮	
10.28.24.12	525400A7798A	VS_SONIC_BUZZNIK_PLUS	spine2	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	🟢	🟢	22-bwester-evps@sonic-vs.2485377892354-6677722504 - evps-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮	
10.28.24.13	52540070D360	VS_SONIC_BUZZNIK_PLUS	leaf1	SONIC 4.0.5-Enterprise_Base	IS-ACTIVE	🟢	🟢	22-bwester-evps@sonic-vs.2485377892354-6677722504 - evps-sonic-vs-virtual	ONBOX	UNASSIGNED	AOS_4.2.0_OB.232	INSTALL	SUCCESS	⋮	
10.28.24.14						—	🔴	Not assigned	ONBOX	UNASSIGNED		INSTALL	⋮		
10.28.24.234	525400A052AA	VS_SONIC_BUZZNIK_PLUS		SONIC 4.0.5-Enterprise_Base	OOS-QUARANTINED	🟢	🔴	Not assigned		UNASSIGNED	AOS_4.2.0_OB.232		⋮		

Agent

9. Confirm the deletion dialog and force the deletion:

Delete System Agent

⚠️ Apstra cannot communicate with this device. It will not be possible to uninstall any agents or revert the device to its Pristine config before deleting. Only select this box if you understand the consequences and cannot reestablish communication with the device.

☒ Force delete?

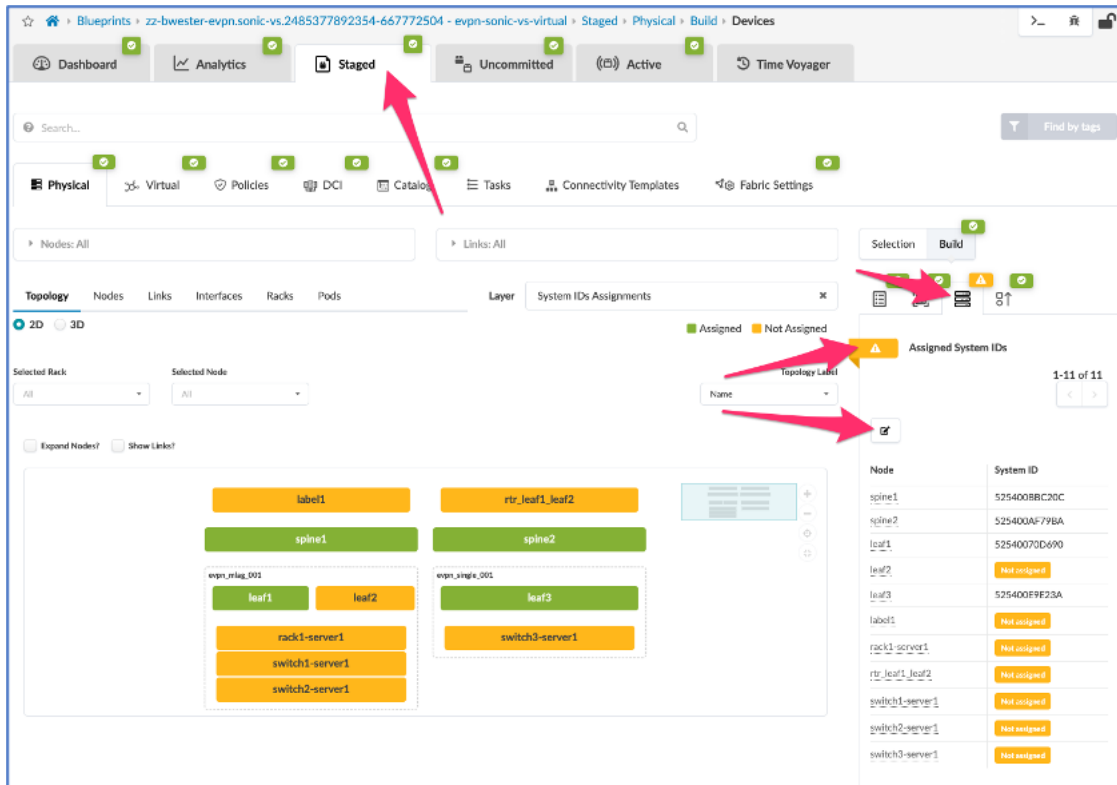
Delete

1.1.3 GUI Method to add the replacement device

Let's now take a look at what would be required to replace the device back into the blueprint.

1. First step is to add the device to Managed Devices. If the device already exists, in Managed Devices you can skip this step. Navigate to Devices -> Managed Devices, and select the "Create Onbox Agents" Icon (SONIC devices use Onbox Agents). You will need to fill in the IP address and Username / Password information:

4. Navigate to Blueprints -> BpID -> Staged -> Build -> Devices and Open “Assigned System IDs”:



5. Select the pencil icon and Click the dropdown next to leaf2 to select the replacement device, then click the “deploy” radio button and finally select the “Update Assignments” selector:

Assign Systems

Query: All 1-11 of 11

Name	Role	Hostname	System ID	Deploy Mode
spine1	Spine	spine1	525400B8C20C (10.28.24.11) - spine1	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
spine2	Spine	spine2	525400AF79BA (10.28.24.12) - spine2	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
leaf1	Leaf	leaf1	52540070D690 (10.28.24.13) - leaf1	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
leaf2	Leaf	leaf2	525400AD52B3 (10.28.24.14) 525400AD52B3 (10.28.24.14)	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
leaf3	Leaf	leaf3	525400E9E23A (10.28.24.15) - leaf3	☒ Deploy ☐ Ready ☐ Drain ☐ Undeploy
label1	Generic System	hostname1	Select...	☐ Deploy ☐ Ready ☐ Undeploy
rack1-server1	Generic System	rack1-server1	Select...	☐ Deploy ☐ Ready

Update Assignments

6. Commit the change by Navigating to the Uncommitted tab, and Click the Commit “Rocket ship”:

Blueprints > zz-bwester-evpn.sonic-vs.2485377892354-667772504 - evpn-sonic-vs-virtual > Uncommitted > Logical Diff

Revert Commit

Dashboard Analytics Staged **Uncommitted** Active Time Voyager

Logical Diff Full Nodes Diff Build Errors Warnings Commit Check

Query: All 1-1 of 1

Page Size: 25

Type	Action	Name
System Node	CHANGED	leaf2

7. Comment and confirm your commit:

Commit changes from Staged to Active?

Information

This action will automatically create a new blueprint revision. This blueprint has 5 out of 5 automatically saved revisions. If the blueprint reaches the maximum amount of revisions to store, this commit will automatically replace the oldest saved revision with this new one.

If you wish to increase the limit for automatically saved revisions, go to [Time Voyager](#) settings. You can also **permanently save older revisions** using the corresponding save button under Time Voyager.

Revision Description (optional):

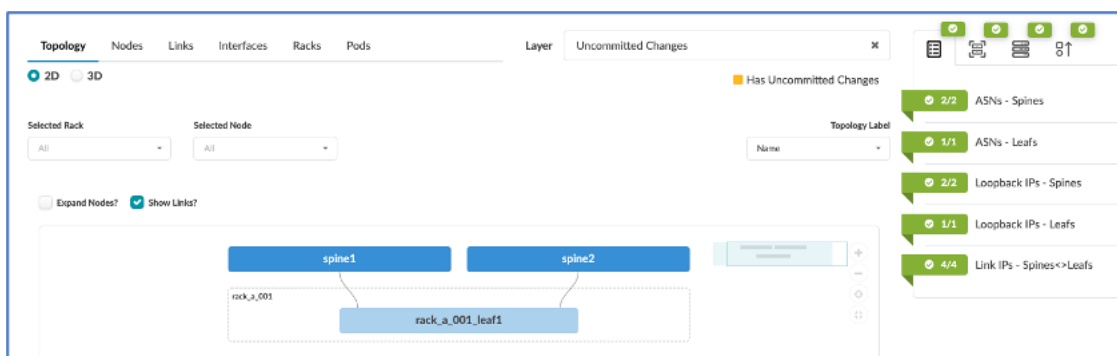
replaced leaf2

Commit

8. At this stage your deployment will take some time to stabilize and the anomalies should eventually converge and disappear.

1.2 Terraform Method to replace a device in Apstra

From a Terraform perspective, replacement of a device is fairly straightforward, and will describe that process with an example terraform configuration of a simple Blueprint:



We will be replacing the device `rack_a_001_leaf1`.

Here is our example Terraform Configuration:

```

1 terraform {
2   required_providers {
3     apstra = {
4       source = "Juniper/apstra"
5     }
6   }
7 }
8 provider "apstra" {
9   url = "https://admin:admin@10.28.24.3:443"
10  tls_validation_disabled = true
11  blueprint_mutex_enabled = false
12  experimental = true #New Apstra 4.2 API
13 }
14 resource "apstra_rack_type" "a_rack" {
15   name = "rack_a"
16   leaf_switches = {
17     leaf1 = {
18       logical_device_id = "slicer-7x10-1"
19       spine_link_speed = "10G"
20       spine_link_count = 1
21     }
22   }
23   fabric_connectivity_design = "l3clos"
24 }
25 resource "apstra_template_rack_based" "a_template" {
26   name = "template_a"
27   asn_allocation_scheme = "unique"

```

```

28 overlay_control_protocol = "evpn"
29 spine = {
30   logical_device_id = "slicer-7x10-1"
31   count = 2
32 }
33 rack_infos = {
34   (apstra_rack_type.a_rack.id) = { count = 1}
35 }
36 }
37 resource "apstra_datacenter_blueprint" "a_blueprint" {
38   name = "blueprint_a"
39   template_id = apstra_template_rack_based.a_template.id
40 }
41 resource "apstra_managed_device_ack" "spine1" {
42   agent_id = "e44df49e-556b-446e-a02f-7fd23804901e"
43   device_key = "525400BBC20C"
44 }
45 resource "apstra_managed_device_ack" "spine2" {
46   agent_id = "b5726b1d-29ba-4ab0-ac76-5def33291cc4"
47   device_key = "525400AF79BA"
48 }
49 resource "apstra_managed_device_ack" "leaf1" {
50   agent_id = "8f448798-c260-46b7-b7fa-46609f0598d4"
51   device_key = "525400E9E2FF"
52 }
53 resource "apstra_datacenter_device_allocation" "spine1" {
54   blueprint_id = apstra_datacenter_blueprint.a_blueprint.id
55   node_name = "spine1"
56   device_key = apstra_managed_device_ack.spine1.device_key
57   deploy_mode = "deploy"
58 }
59 resource "apstra_datacenter_device_allocation" "spine2" {
60   blueprint_id = apstra_datacenter_blueprint.a_blueprint.id
61   node_name = "spine2"
62   device_key = apstra_managed_device_ack.spine2.device_key
63   deploy_mode = "deploy"
64 }
65 resource "apstra_datacenter_device_allocation" "leaf1" {
66   blueprint_id = apstra_datacenter_blueprint.a_blueprint.id
67   node_name = "rack_a_001_leaf1"
68   device_key = apstra_managed_device_ack.leaf1.device_key
69   deploy_mode = "deploy"
70 }

```

As you can see this is a very simple Terraform configuration.

We assume the terraform configuration is applied to the Apstra server, and all of the Resources for the blueprint are Allocated and Committed.

Steps needed to replace a device using the Terraform Workflow:

1. Remove the resource for the device from the Terraform configuration, in this case we will be removing the following and applying the configuration.

```

1 resource "apstra_datacenter_device_allocation" "leaf1" {
2   blueprint_id = apstra_datacenter_blueprint.a_blueprint.id
3   node_name = "rack_a_001_leaf1"
4   device_key = apstra_managed_device_ack.leaf1.device_key
5   deploy_mode = "deploy"
6 }

```

By removing this configuration and `terraform apply` we remove the `rack_a_001_leaf1` from the blueprint and prepare it for removal from Managed Devices. You cannot remove a device from Apstra without first removing it from the Appropriate Blueprint.

Here is the result of the `terraform apply`

```

1 bwester@bwester-mbp terraform % terraform apply
2 apstra_datacenter_device_allocation.leaf1: Refreshing state...
3 apstra_managed_device_ack.spine2: Refreshing state...
4 apstra_managed_device_ack.leaf1: Refreshing state...
5 apstra_managed_device_ack.spine1: Refreshing state...
6 apstra_rack_type.a_rack: Refreshing state... [id=wjdosvpvs8qdkyec2slrw]

```

```

7  apstra_template_rack_based.a_template: Refreshing state... [id=3fcd246d-f7eb-4850-ba18-
   bd8429aa3c8d]
8  apstra_datacenter_blueprint.a_blueprint: Refreshing state... [id=9c2404f6-a875-4d29-9add
   -919601601b04]
9  apstra_datacenter_device_allocation.spine2: Refreshing state...
10 apstra_datacenter_device_allocation.spine1: Refreshing state...
11
12 Terraform used the selected providers to generate the following execution plan. Resource
   actions are indicated with the following symbols:
13   - destroy
14
15 Terraform will perform the following actions:
16
17   # apstra_datacenter_device_allocation.leaf1 will be destroyed
18   # (because apstra_datacenter_device_allocation.leaf1 is not in configuration)
19   - resource "apstra_datacenter_device_allocation" "leaf1" {
20     - blueprint_id = "9c2404f6-a875-4d29-9add-919601601b04" -> null
21     - deploy_mode = "deploy" -> null
22     - device_key = "525400E9E23A" -> null
23     - device_profile_node_id = "sfBQVnyC5MBMtWoYLPu" -> null
24     - initial_interface_map_id = "VS_SONiC_BUZZNIK_PLUS__slicer-7x10-1" -> null
25     - interface_map_name = "VS_SONiC_BUZZNIK_PLUS__slicer-7x10-1" -> null
26     - node_id = "v4U3_ek6bH3t_ppYEXA" -> null
27     - node_name = "rack_a_001_leaf1" -> null
28   }
29
30 Plan: 0 to add, 0 to change, 1 to destroy.
31
32 Do you want to perform these actions?
33   Terraform will perform the actions described above.
34   Only 'yes' will be accepted to approve.
35
36   Enter a value: yes
37
38 apstra_datacenter_device_allocation.leaf1: Destroying...
39 apstra_datacenter_device_allocation.leaf1: Destruction complete after 4s
40
41 Apply complete! Resources: 0 added, 0 changed, 1 destroyed.

```

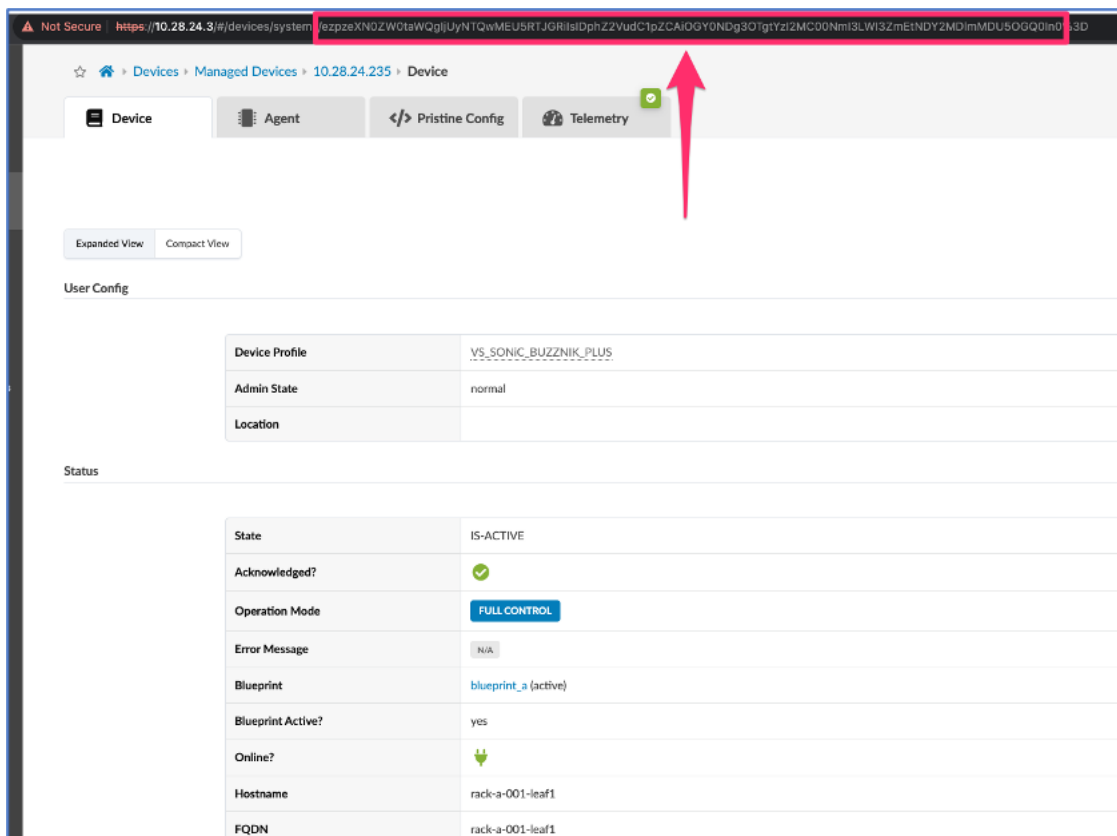
2. Once this step is complete you will need to commit the blueprint changes.
3. We now will update the terraform configuration with the new information for the replacement device.
The specific area of the Terraform configuration we are interested in is the following snippet:

```

1  resource "apstra_managed_device_ack" "leaf1" {
2    agent_id = "8f448798-c260-46b7-b7fa-46609f0598d4"
3    device_key = "525400E9E2FF"
4  }

```

Replace the `device_key` with the new Key value. Replace the `agent_id` with the new Agent ID value. This value can be derived from navigating to the UI via Devices -> Managed Devices -> `device ip` -> Device Copy the encoded Device Key and AgentID from the URL string.



Base64 Decode the information:

```
1 pbpaste | base64 --decode
2 { :system-id "525400E9E2FF", :agent-id "8f448798-c260-46b7-b7fa-46609f0598d4%
```

You can also derive the agent-id from a data source, or from swagger, or from the terraform resource which created it.

4. Once you have replaced the information into the terraform configuration, apply the new Terraform.

```
1 bwester@bwester-mbp terraform % terraform apply
2 apstra_managed_device_ack.spine1: Refreshing state...
3 apstra_managed_device_ack.spine2: Refreshing state...
4 apstra_rack_type.a_rack: Refreshing state... [id=wjdsvpvs8qdkyec2slrw]
5 apstra_template_rack_based.a_template: Refreshing state... [id=3fcd246d-f7eb-4850-ba18-
6 bd8429aa3c8d]
7 apstra_datacenter_blueprint.a_blueprint: Refreshing state... [id=9c2404f6-a875-4d29-9add
8 -919601601b04]
9 apstra_datacenter_device_allocation.spine2: Refreshing state...
10 apstra_datacenter_device_allocation.spine1: Refreshing state...
11
12 Terraform used the selected providers to generate the following execution plan. Resource
13 actions are indicated with the following symbols:
14 + create
15
16 Terraform will perform the following actions:
17
18 # apstra_datacenter_device_allocation.leaf1 will be created
19 + resource "apstra_datacenter_device_allocation" "leaf1" {
20   + blueprint_id = "9c2404f6-a875-4d29-9add-919601601b04"
21   + deploy_mode = "deploy"
22   + device_key = "525400E9E2FF"
23   + device_profile_node_id = (known after apply)
24   + initial_interface_map_id = (known after apply)
25   + interface_map_name = (known after apply)
26   + node_id = (known after apply)
27   + node_name = "rack_a_001-leaf1"
28 }
29
30 # apstra_managed_device_ack.leaf1 will be created
31 + resource "apstra_managed_device_ack" "leaf1" {
```

```
29     + agent_id = "8f448798-c260-46b7-b7fa-46609f0598d4"
30     + device_key = "525400E9E2FF"
31     + system_id = (known after apply)
32   }
33
34   Plan: 2 to add, 0 to change, 0 to destroy.
35
36   Do you want to perform these actions?
37     Terraform will perform the actions described above.
38     Only 'yes' will be accepted to approve.
39
40   Enter a value: yes
41
42   apstra_managed_device_ack.leaf1: Creating...
43   apstra_managed_device_ack.leaf1: Creation complete after 1s
44   apstra_datacenter_device_allocation.leaf1: Creating...
45   apstra_datacenter_device_allocation.leaf1: Creation complete after 7s
46
47   Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
```

Now that you have changed the system you will need to commit the changes to Apstra.

In these examples, we did all of the work to clean any existing configuration from the new device, boot the device, and install the Apstra Agent on the device. For example in the SONIC use case, you may be required to remove any existing configuration from the device, here is an example:

```
1   root@sonic:/home/admin# rm /etc/sonic/config_db.json
2   root@sonic:/home/admin# config-setup factory
3   root@sonic:/home/admin# reboot
```

1.3 Conclusion

Device replacement in Apstra is a straightforward, easy process. This article demonstrates two methodologies: replacement in the Apstra UI, and replacement using Terraform.

1.4 Acknowledgements

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